

The recovered contraction maps one unit ball onto the other

MathlibAnnex.PluckerRecovery.LimitCertificate.image_unitBall_eq

theorem

Upgrades ball inclusion to equality by strict compact-subset volume comparison.

Statement

Let $n = m + 1 > 0$ and $V = \mathbb{R}^n$ with the reference sup norm and standard Lebesgue measure λ . The continuous norms p, q have fixed positive comparison constants

$$a_p \|x\|_\infty \leq p(x) \leq b_p \|x\|_\infty, \quad a_q \|x\|_\infty \leq q(x) \leq b_q \|x\|_\infty.$$

Set $K_p = \{x : p(x) \leq 1\}$, $K_q = \{x : q(x) \leq 1\}$ and $v_p = \lambda(K_p)$, $v_q = \lambda(K_q)$, as positive finite real numbers. Let $L : V \rightarrow V$ be a real linear map with the following properties: $q(Lx) \leq p(x)$ for every x , $L(K_p) \subseteq K_q$, $v_p |\det L| = v_q$, $\det L \neq 0$, and L is bijective. Then

$$L(K_p) = K_q.$$

Assumptions

The two norm models and one already fixed real linear map L with the seven displayed certificate properties are inputs. In particular, its ball inclusion and determinant-volume equation must concern the same L . Equality of Plücker bodies is not an input to this theorem.

Conclusion

Every image of a p -unit-ball vector is in K_q , and every point in K_q is such an image.

Proof route

Compute the image volume for the same map and exclude a proper compact subset of the target unit ball.

▼ Proof steps

1. **Compute the volume of this same linear image.** The change-of-variables formula gives

$$\begin{aligned} \lambda(L(K_p)) &= |\det L| \lambda(K_p) \\ &= |\det L| v_p \\ &= v_q = \lambda(K_q). \end{aligned}$$

Here the first equality is the linear image-volume formula, used with Lebesgue measure λ , the continuous linear map L , and the set K_p . The second uses $v_p = \lambda(K_p)$; the third uses the assumed identity $v_p |\det L| = v_q$. This is the image-volume calculation for the map in the supplied certificate.

All volumes in this calculation are finite: K_p, K_q are compact, L is continuous, and Lebesgue measure is finite on compact sets. We therefore identify these finite measures with the same nonnegative real numbers throughout. The resulting equality of image and target measures concerns the same $L(K_p)$ and K_q that occur in the inclusion hypothesis.

2. **Rule out a proper compact subset.** Continuity of L makes $L(K_p)$ compact. If its known inclusion in K_q were proper, strict measure loss inside a seminorm ball would give

$$\lambda(L(K_p)) < \lambda(K_q).$$

The inputs at this use are Lebesgue measure, continuous seminorm q , compactness of the subset, inclusion and inequality of the sets. The output contradicts Step 1. Thus $L(K_p) = K_q$. Equal measures of arbitrary sets alone would not justify equality.

The norm comparisons provide compact unit balls and positive finite volumes. Every volume used in the proof is finite; no infinite measure is treated as a real number.

Main citations

- Exact declaration and source
- linear image-volume calculation
- certificate measure equality
- Haar linear-image provider
- strict measure loss inside a seminorm ball

Lean source signature (exact)

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-- Inclusion and exact measure force equality: strict containment loses measure. -/
theorem image_unitBall_eq : C.linearMap '' MX.closedUnitBall = MY.closedUnitBall
```

In the source	Mathematical meaning
<code>Fin (m + 1) → ℝ; →L[ℝ]</code>	The reference $V = \mathbb{R}^n$, $n = m + 1 > 0$, with sup norm; arrows mean continuous real linear maps. Source indices $0, \dots, n - 1$ correspond to formula indices $1, \dots, n$.
<code>MX MY : EquivalentSeminorm (Fin (m + 1) → ℝ); MX.p; MY.p</code>	The two continuous norms p, q and their positive comparison constants. The applications <code>MX.p x</code> and <code>MY.p x</code> mean $p(x), q(x)$.
<code>MX.closedUnitBall; MY.closedUnitBall; MX.closedUnitBallVolume; MY.closedUnitBallVolume</code>	The sets K_p, K_q and their real Lebesgue volumes v_p, v_q . These finite nonnegative measures are read as real numbers.

In the source	Mathematical meaning
<code>C : LimitCertificate MX MY; C.linearMap</code>	The parameter C is supplied in the surrounding declaration context. It consists of the already fixed real linear map L and exactly the six properties read in the following field rows.
<code>C.linearMap '' MX.closedUnitBall</code>	The set $\{Lx : x \in V, p(x) \leq 1\}$. The double apostrophe means set image.
<code>C.linearMap '' MX.closedUnitBall = MY.closedUnitBall</code>	The whole conclusion $L(K_p) = K_q$, as sets rather than just as measures.
<code>volume; toReal</code>	volume is Lebesgue measure with values in $[0, \infty]$. For a finite value, <code>.toReal</code> reads the same nonnegative number as a real number. Here compactness establishes finiteness before this operation is used.
<code>C.seminorm_linearMap_le</code>	For every x , $q(Lx) \leq p(x)$, about the supplied L .
<code>C.image_closedUnitBall_subset</code>	$L(K_p) \subseteq K_q$, the inclusion to be strengthened.

In the source	Mathematical meaning
<code>C.closedUnitBallVolume_mul_abs_det</code>	$v_p \det L = v_q$, for the determinant of this same map.
<code>C.det_ne_zero; C.injective; C.surjective</code>	The other three inputs are $\det L \neq 0$, injectivity and surjectivity. Together with <code>linearMap</code> these exhaust the seven fields of the supplied certificate.

The final equality is the conclusion about this supplied L . The `''` notation means image of a set, not a derivative. The certificate assumptions come from the separately linked surrounding source, rather than from an equality-of-bodies premise.

Surrounding assumptions and aliases.

▼ Exact content identity

Declaration: `MathlibAnnex.PluckerRecovery.LimitCertificate.image_unitBall_eq`

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